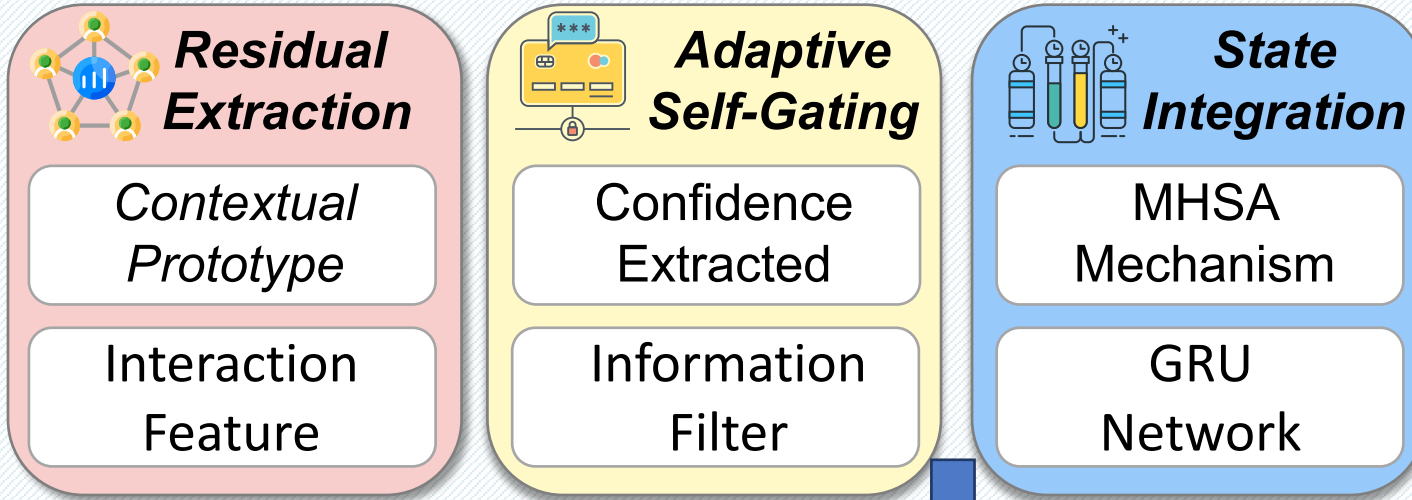
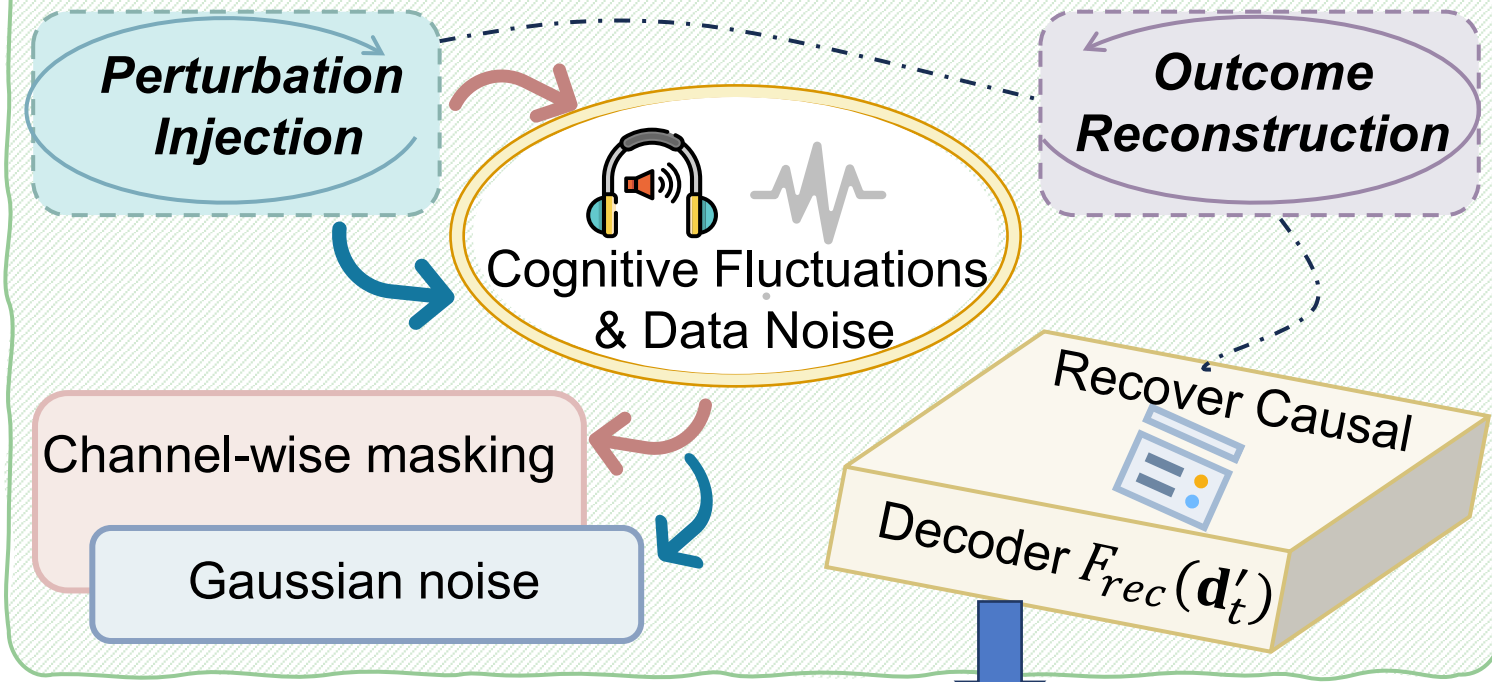


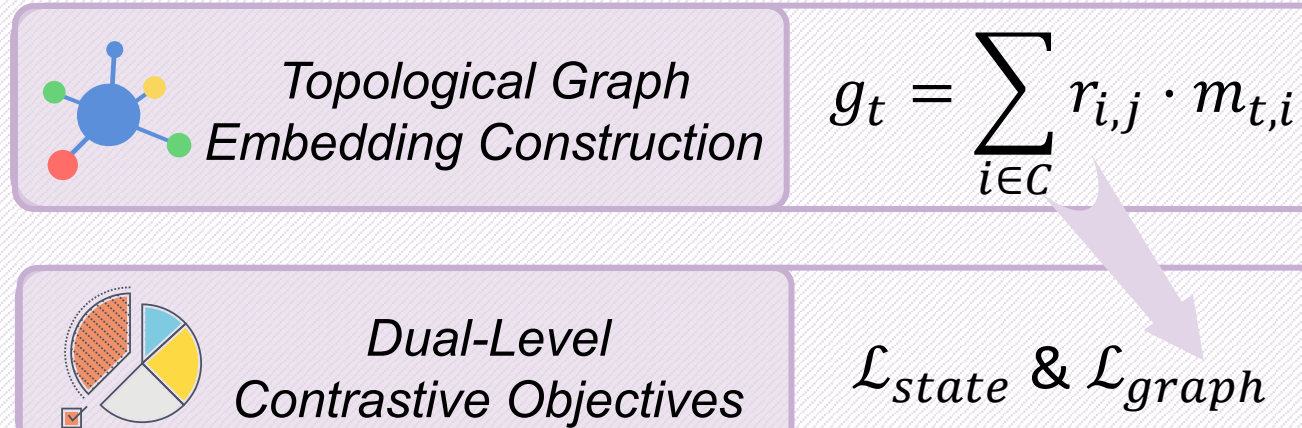
(a) Gated Residual Interaction Encoding



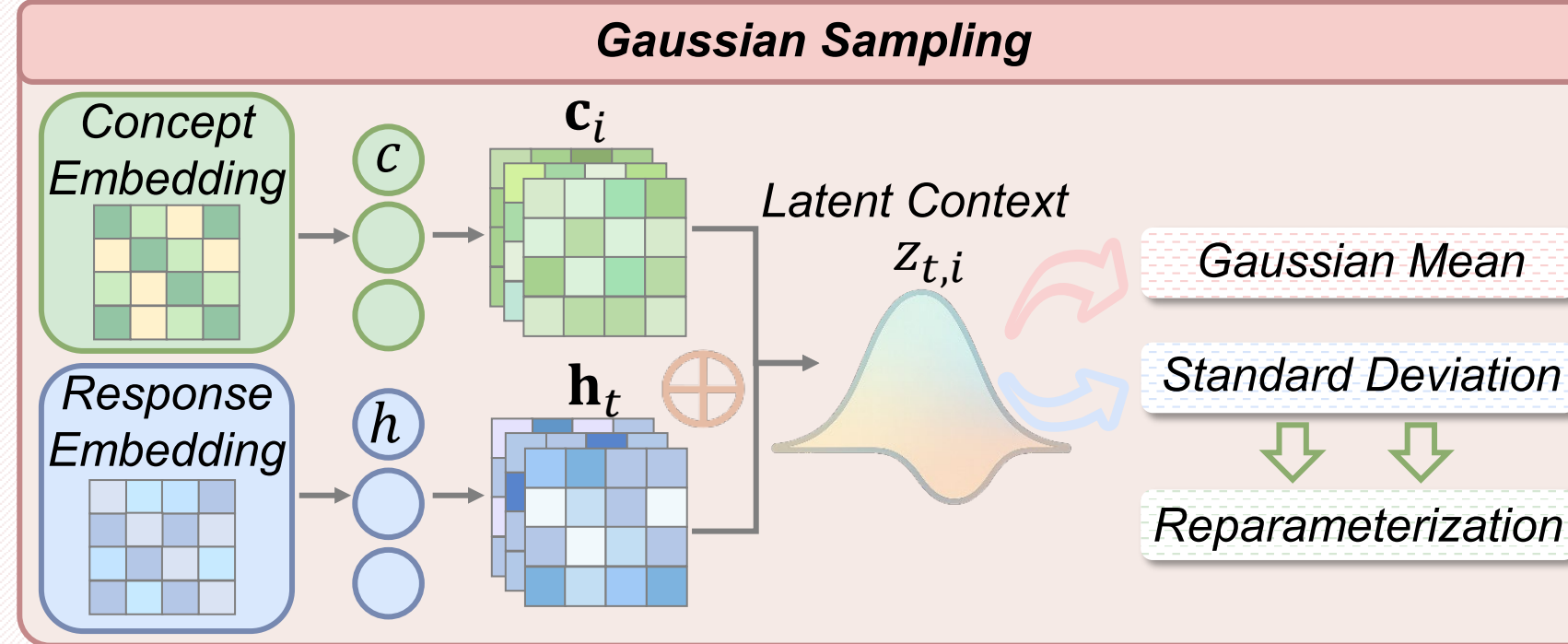
(b) Perturbation-Reconstruction Auxiliary Task



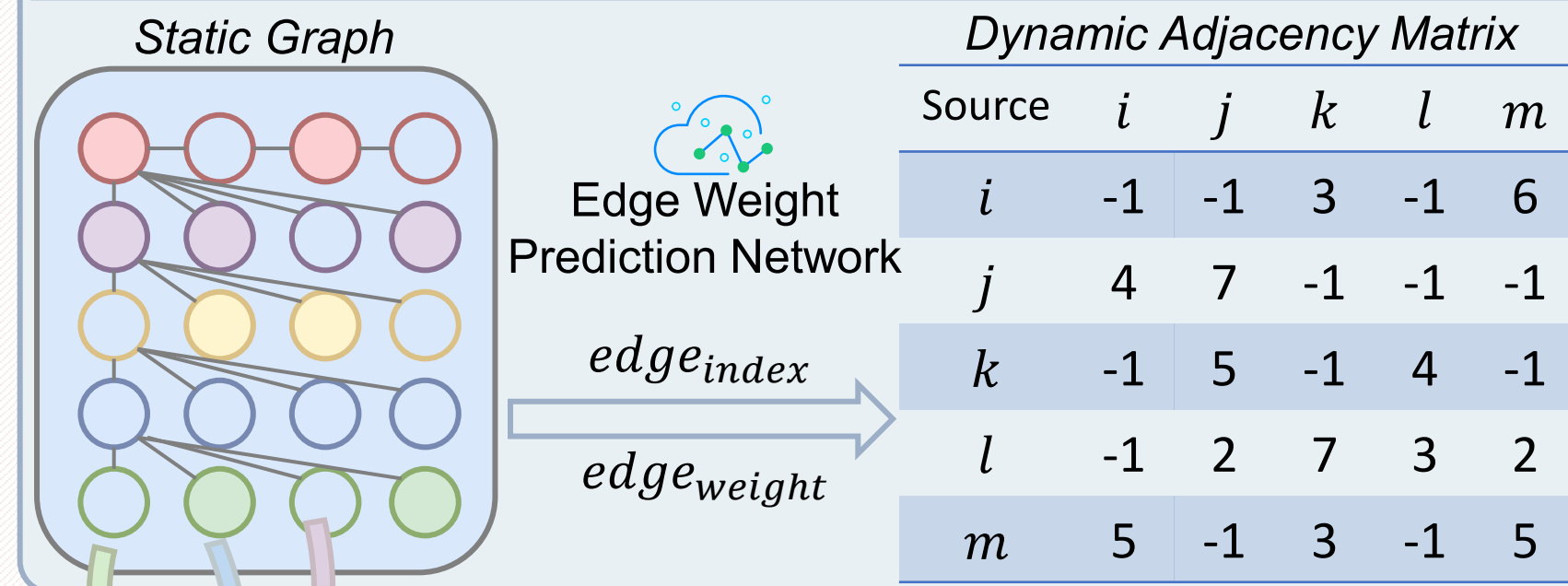
(e) Path-Aware Contrastive Learning



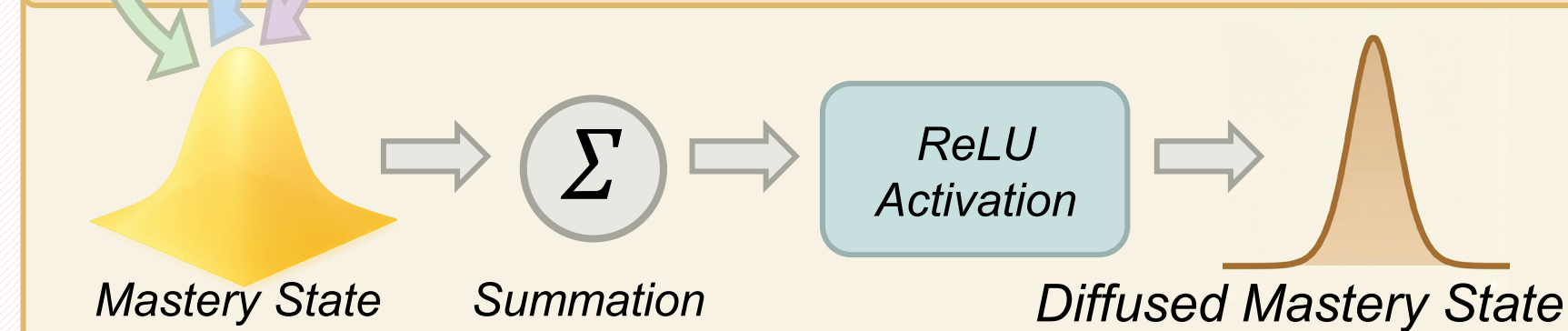
(c) Gaussian Graph Diffusion Encoder



Dynamic Graph Construction

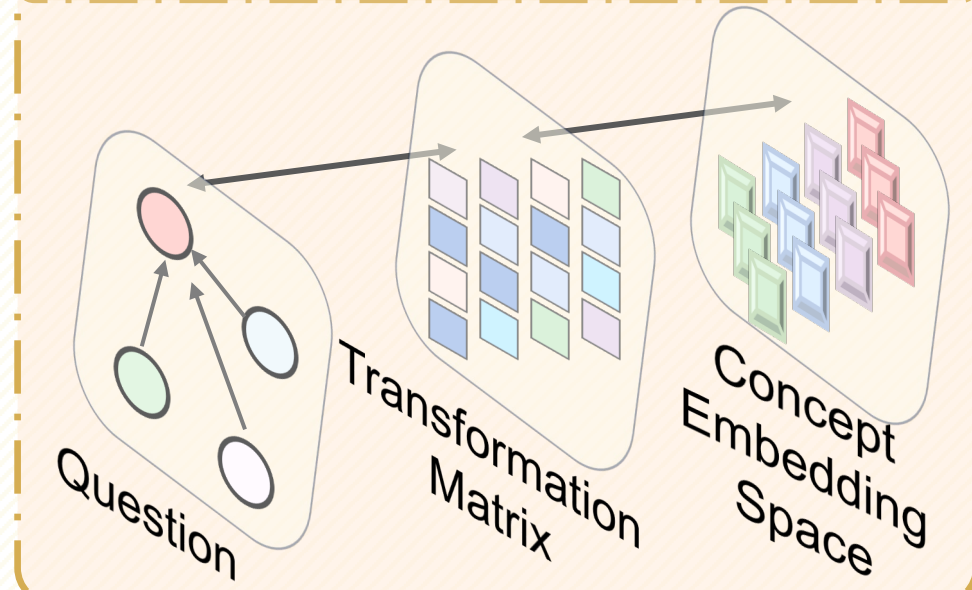


Gaussian propagation

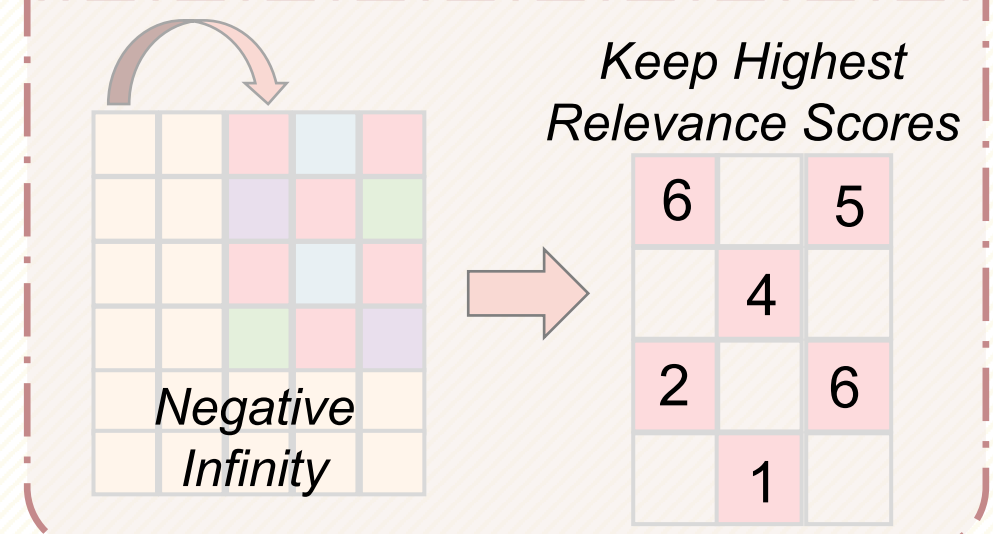


(d) Structure-Aware IRT Prediction

Concept Relevance Alignment



Top-K Sparse Focusing



IRT-Based Probability Estimation

